الإسم والشهرة : .....

رقم المرشح : .....

الجمهورية اللبنانية
وزارة التربية والتعليم العالي
المديرية العامة للتعليم المهني والتقني

الإمتحانات الرسمية لشهادة الإمتياز الفني
لدورة عام ٢٠١٩ الأولى

الإختصاص والرمز : الإلكترونيك (٢٠٢٠٣)
المادة : الإلكترونيك الرقمي
المستندات المسموح بها : آلة حاسبة غير مبرمجة

المعدل : (١٠)
المدة : ساعتان

Part A : MP 6809

Treat two questions only: (5 pts /question)

A1- Write the program that adds the contents of two lists of unsigned numbers, byte by byte, and store the result in a third list.

The first list starts at $0000 and ends at $0004
The second list starts at $0010 and ends at $0014
The third list starts at $0020 and ends at $0024

Example:

List 1 Address	Content
$0000	05
$0001	87
$0002	78
$0003	EA
$0004	F3

+

List 2 Address	Content
$0010	08
$0011	11
$0012	13
$0013	87
$0014	06

=

List 3 Address	Content
$0020	0D
$0021	98
$0022	8B
$0023	71
$0024	F9

A2- S1 and S0 are two switches connected, via port A (ORA) of PIA 6821, to MP 6809 and L0 to L7 are 8 LEDs connected, via port B (ORB). Draw the necessary connection circuit and write the corresponding program that allows to carry out the below table:

S1	S0	L7	L6	L5	L4	L3	L2	L1	L0
0	0	1	1	1	1	0	0	0	0
0	1	0	0	1	1	1	1	0	0
1	0	0	0	0	0	1	1	1	1
1	1	1	1	1	1	1	1	1	1

A3- a- Give the result of each of the following instructions and indicate the used addressing mode with the following initial conditions: A=$20, B =$21, C =1 and X = $1000.

Instructions	Acc A	Acc B	C (carry)	X (Index)	Addressing mode
ADDA #$30					
ROLA					
ASRA					
SUBA #$25					
ABX					
MUL					
LDX #$3636					
EORB #$52					
TFR B, A					
ADCB #$23					

b- Name 5 internal registers of MP 6809 and specify the role and the size of each one (8 bits or 16 bits).

دورة C.19 الأولى - الإلكترونيك الرقمي "ميكرو" T.S.

PART A: 6809.

A1.

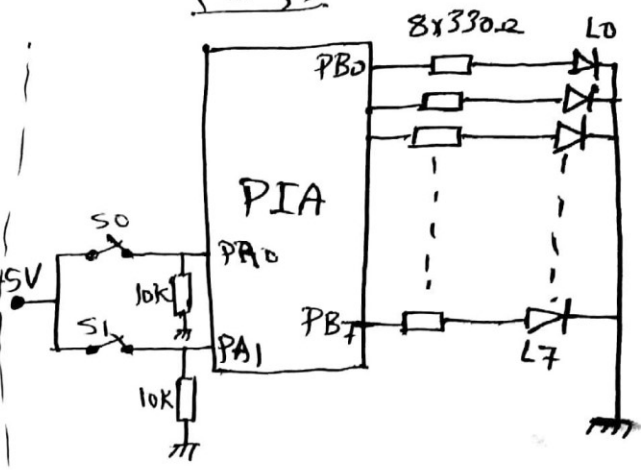
```
        ORG $FC00
        LDX #$0000
        LDY #$0010
        LDU #$0020
L1:     CLRA
        ADDA ,X+
        ADDA ,X+
        STA ,U+
        CMPX #$0005
        BNE L1
        END
```

A2.

البرنامج:

```
PORTA EQU $8000
CRA EQU $8001
PORTB EQU $8002
CRB EQU $8003
ORG $FC00
CLRCRA
LDA #%11111100
STA PORTA
CLR CRB
LDA #$FF
STA PORTB
LDA #$04
STA CRA
STA CRB
LOOP:
LDA PORTA
BEQ L0
CMPA #$01
BEQ L1
CMPA #$02
BEQ L2
:
:
LDA #$FF
STA PORTB
BRA LOOP
L0:
LDA #%11110000
STA PORTB
BRA LOOP
L1:
LDA #%00111100
STA PORTB
BRA LOOP
L2:
LDA #%00001111
STA PORTB
BRA LOOP
END
```

رسم الدارة:



Scanned with CamScanner

A3

9-

Instruction	AccA	AccB	C	X	Addressing
ADDA #$30	$50	$21	0	$1000	Immediate
ROL A	$A0	$21	0	//	Inherent
ASRA	$D0	$21	0	//	//
SUBA #$25	$AB	$21	0	//	Immediate
ABX	//	$21	//	$1021	Inherent
MUL	$16	$0B	0	//	Inherent
LDX #$3636	//	//	//	$3636	Immediate
EORB #$52	//	$58	//	//	//
TFR B, A	$58	$58	//	//	Immediate
ADCB #$23	//	$7B		//	Immediate

~~[illegible]~~

b- 1. Acc A+B : each is 8 bit register used for arithmatic + logic operations.

2. X+Y : each is 16 bit register used as index registers for pointing data in index mode operations and other uses.

3. S+U : each is 16 bit register used a stack pointers used by the processor or the user during subroutine calls + interrupts.

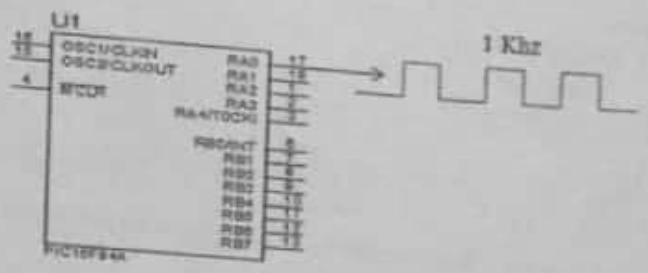
4. PC : program counter is a 16 bit register used to point the address of the next instruction

5. CCR : condition code register is an 8 bit register that defines the state of the processor at any given time. Each bit acts as a flag for an event or state change.

Part B : PIC 16F84A

Treat two questions only: (5 pts /question)

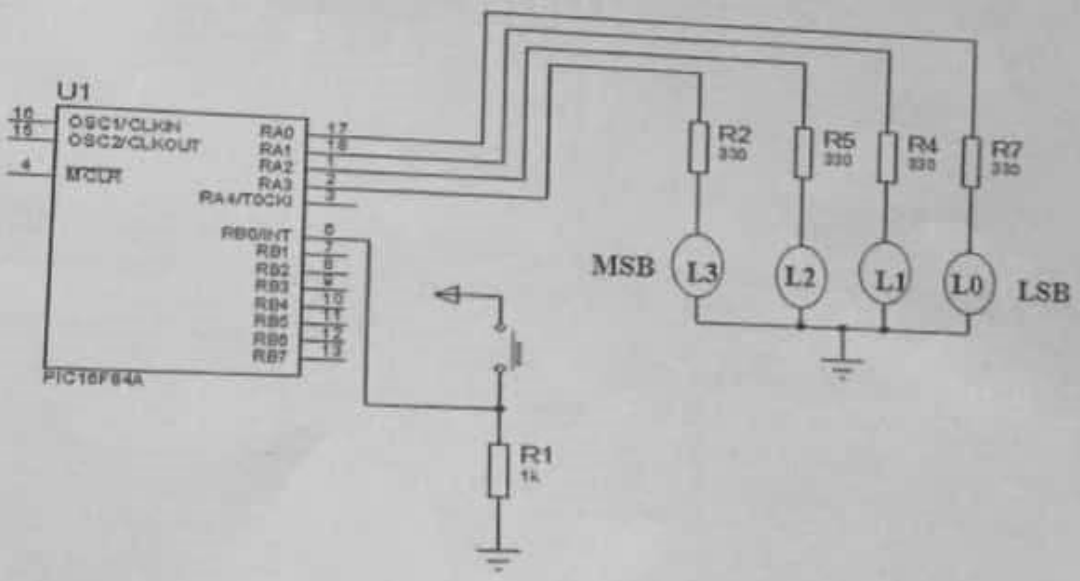
B1- By referring to the circuit of the figure below, write the program that allows to generate, by using the TIMER 0, a square waveform signal with 50% of duty cycle and 1 Khz of frequency on the pin RA0. The frequency of quartz is 8 MHz.



B2- Write the program that allows to display at the port B, sequentially, the contents of the table below, with a delay, at you choice, separated two consecutives displays:

PB7	PB6	PB5	PB4	PB3	PB2	PB1	PB0
1	1	1	1	1	1	1	0
1	1	1	1	1	1	0	0
1	1	1	1	1	0	0	0
1	1	1	1	0	0	0	0
1	1	1	0	0	0	0	0
1	1	0	0	0	0	0	0
1	1	0	0	0	0	0	0
1	0	0	0	0	0	0	0

B3- By referring to the circuit of the figure below, write the program that allows to count the number of active falling edge at the interrupt input RB0 / INT, and displays the result on port A.



PartB: PIC 16F84A

B1.

```
list p=16f84a
include <P16F84A.INC>

org 0x00
goto start

start bsf STATUS, RP0          ;switch to Bank 1
bsf STATUS, RP0                ;switch to Bank 1; OPTION
                               register is at Bank 1

movlw 0x00
movwf TRISA                    ;PORTA all output
movlw 27h                      ;binary 00100111 - maximum
movwf OPTION_REG               prescale, assigned to TMR0
bcf STATUS, RP0                ;switch to Bank 0
clrf TMR0                      ;initialize the timer to zero

 main                          ;----------------main routine
movf TMR0,0                    ;move contents of TMR0 to W
andlw 0x08                     ;mask so that only bit 0 of W is
movwf PORTA                    moved
goto main                      ;to PORTA
end
```

PartB: PIC 16F84A

B2.

```
list p=16f84a
include <P16F84A.INC>


org 0x00
goto start

start bsf STATUS, RP0      ;switch to Bank 1
bsf STATUS, RP0            ;switch to Bank 1; OPTION register is at Bank 1

movlw 0x00
movwf TRISB                ;PORTB all output
bcf STATUS, RP0            ;switch to Bank 0
movlw b'11111110'          ;set initial value
movwf PORTB                ;move to PORTB


                           ;---------------main routine

 main
nop                        ;delay 1 cycle
nop                        ;delay 1 cycle
bcf STATUS,c               ;clear Carry bit
rlf PORTB,1                ;rotate left data in PORTB
goto main                  ;repeat
end
```

PartB: PIC 16F84A

B3.

```
list p=16f84a
include <P16F84A.INC>
RES_VECT  code 0x0000
goto    start
INT_VECT  code  0x0004
goto    isr
org 0x00
start
bsf STATUS, RP0
clr TRISA
clr TRISB
bsf TRISB,0
bcf STATUS, RP0
movlw b'10010000'
movwf INTCON
goto main

isr
bcf INTCON, GIE
bcf PORTA,0 ;
bcf INTCON,INTF
bsf INTCON, GIE
retfie

main
Incf PORTA,0
goto main

end
```

```
; processor reset vector
; go to beginning of program
; interrupt vector
; go to interrupt service routine


;switch to Bank 1;
;PORTA all output

;RP0 input
;switch to Bank 0

;Global interrupt enabled,
External interrupt enabled


;Disable all interrupts
;clear RA.0
;Clear interrupt flag bit
;Enable all interrupts on ;exit


;main routine prog
; increment PORTA
;loop
```

APPENDIX

Instruction set of MP 6809

ADCA, ADCB, ADDA, ADDB, ADDD, ANDA, ANDB, ANDCC, ASLA, ASLB, ASL, ASRA, ASRB, ASR, BITA, BITB, CLRA, CLRB, CLR, CMPA, CMPB, CMPD, CMPU, CMPS, CMPX, CMPY, COMA, COMB, COM, CWAI, DAA, DECA, DECB, DEC, EORA, EORB, EXG, INCA, INCB, INC, JMP, JSR, LDA, LDB, LDD, LDS, LDU, LDX, LDY, LEAS, LEAU, LEAX, LEAY, LSLA, LSLB, LSL, LSRA, LSRB, LSR, MUL, NEGA, NEGB, NEG, NOP, ORA, ORB, ORCC, PSHS, PSHU, PULS, PULU, ROLA, ROLB, ROL, RORA, RORB, ROR, RTI, RTS, SBCA, SBCB, SEX, STA, STB, STD, STS, STU, STX, STY, SUBA, SUBB, SUBD, SWI, SWI2, SWI3, SYNC, TFR, TSTA, TSTB, TST.

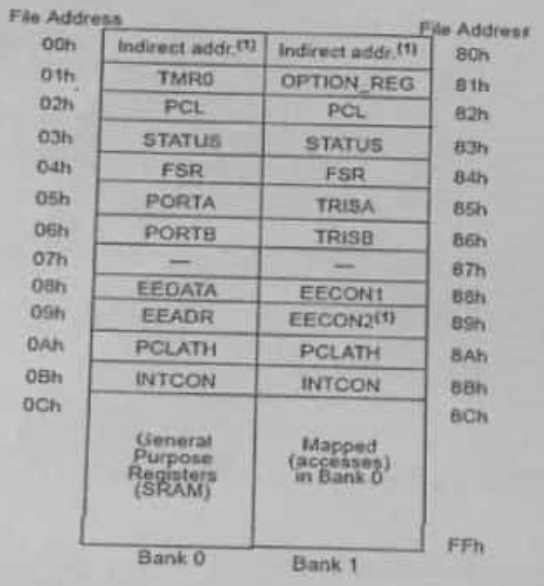
PIA 6821:

ORA and DDRA = $8000	**CRA = $8001**
ORB and DDRB = $8002	**CRB = $8003**

Instruction set of PIC 16F84A

ADDWF, ANDWF, CLRF, CLRW, COMF, DECF, DECFSZ, INCF, INCFSZ, IORWF, MOVF, MOVWF, NOP, RLF, RRF, SUBWF, SWAPF, XORWF, BCF, BSF, BTFSC, BTFSS, ADDLW, ANDLW, CALL, CLRWDT, GOTO, IORLW, MOVLW, RETFIE, RETLW, RETURN, SLEEP, SUBLW, XORLW.

Internal architecture of PIC16F84A



PS2:PS0: Prescaler Rate Select bits

Bit Value	TMR0 Rate	WDT Rate
000	1 : 2	1 : 1
001	1 : 4	1 : 2
010	1 : 8	1 : 4
011	1 : 16	1 : 8
100	1 : 32	1 : 16
101	1 : 64	1 : 32
110	1 : 128	1 : 64
111	1 : 256	1 : 128

Time = {(256 – TMR0) /(Fosc/4)}x Prescaler

OPTION_REG Register

RBPU	INTEDG	T0SC	T0SE	PSA	PS2	PS1	PS0

NTCON Register

GIE	EEIE	T0IE	INTE	RBIE	T0IF	INTF	RBIF